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Leveraging Data Analytics to Design and Drive an Integrated Flaring Reduction Strategy at Tangguh LNG

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Abstract

This paper presents a data-driven approach to flaring reduction at the Tangguh Liquefied Natural Gas (LNG) facility by embedding analytics and control valve modeling into operational strategy, enabling targeted interventions across routine, non-routine, safety, and start-up/shutdown events. The methodology begins with source-level quantification using historian data from 74 flare-connected valves across three LNG trains, where valve openings, pressure differentials, and gas properties were processed via the control valve equation to estimate individual flow contributions. These contributions were categorized by operational source and benchmarked against Minimum Technical Limits (MTLs), which were derived from historical low points during safe, stable operation and served as reference baselines for each category. A real-time flaring dashboard was developed to visualize daily volumes and identify high-emitting sources ("monsters"), which significantly improved operational accountability and focus. Based on this framework, three strategic levers were implemented to reduce flaring: optimizing LNG loading and dryer bed transitions during routine operation, rationalizing purge gas consumption, and harmonizing start-up sequences across shifts. Collectively, these measures yielded a technical reduction potential of 1,963 MMSCF/year—equivalent to a 53% cut in total flare emissions and valued at approximately USD 13.74 million per year, assuming a gas heating value of 1,000 Btu/scf and a market price of USD 7.0/MMBtu. By Jan - June 2025, 1,141 MMSCF/year of this target had been realized (58%), translating to USD 7.99 million in economic value and 62,192 tCO₂e in avoided emissions. This paper offers a fresh perspective on flaring reduction by embedding data analytics at the core of strategy development; it moves beyond traditional engineering reviews by introducing MTL baselines, real-time tracking, and structured source classification. The result is a replicable, data-backed framework that enables engineers to drive impactful emission cuts with greater precision and clarity, particularly in complex LNG facilities where instrumentation is often limited to flare headers. With no requirement for physical hardware upgrades, this approach demonstrates that significant emissions savings can be achieved through better visibility, smarter prioritization, and procedural alignment—making it a practical reference for emissions management in similar high-throughput energy operations.

Introduction

Tangguh LNG is located in the remote Bintuni Bay of West Papua, Indonesia. With a nameplate capacity of 11.4 million tonnes per annum across three liquefaction trains, the facility plays a key role in supporting both domestic energy supply and long-term export commitments. As with many integrated LNG operations, flaring remains a vital safeguard mechanism during plant upsets and abnormal situations. However, in steady-state operations, minimizing avoidable flaring has become a strategic imperative—both from an environmental and operational excellence standpoint.

Industry-wide, reducing flaring has gained renewed urgency. The International Association of Oil and Gas Producers (IOGP) reported that in 2023 ([IOGP Report, 2023](#)), the average flaring intensity among its participating companies reached 8.8 tonnes of hydrocarbon flared per thousand tonnes of hydrocarbon produced. This figure serves not only as a performance benchmark but also as a call to action for operators to reevaluate the systems, behaviors, and practices that drive flaring in their operations.

At Tangguh, flaring visibility was historically limited to aggregated readings at the flare knock-out drums (KODs). While sufficient for compliance tracking, this approach lacked the granularity to attribute flaring to specific process equipment or events. As a result, the plant's ability to proactively manage and reduce flaring was constrained.

This paper outlines the design and deployment of a data-driven flaring management strategy at Tangguh LNG, centered on control valve-level analytics, operational categorization of flaring sources, and real-time visualization. It introduces the concept of **Minimum Technical Limits (MTLs)** to define the lowest safe flaring baseline for different categories of operation, and presents a dashboard-driven workflow for continuous monitoring and intervention.

Together, these elements form an integrated framework that enables Tangguh LNG to shift from reactive to proactive flaring management—unlocking emission reduction, value recovery, and enhanced operational discipline in a scalable, replicable manner.

Statement of Problem and Technical Context

Modern LNG facilities are equipped with multiple flare systems designed to ensure safe release of gases during abnormal conditions, maintenance, and operational transitions. However, while these systems play a vital safety role, they also represent a significant source of routine and non-routine emissions—many of which are preventable. Optimizing flare performance therefore requires not only operational discipline, but also precise visibility into where, when, and why flaring occurs.

At the subject facility, flare metering has historically been limited to a small number of bulk flowmeters located at the main flare knock-out drums (KODs) of each system—namely the Wet, Dry, and Tankage headers. These red-node points, shown in [Figure 1](#), represent the only locations where flare volumes are continuously measured. While adequate for high-level monitoring and regulatory reporting, this setup offers no visibility into individual equipment behavior or process deviations that may drive excess flaring upstream.

The flare network itself is complex, comprising dozens of control valves, vent lines, and relief devices across three production trains. Each of these elements may discharge into the same header, yet the system lacks instrumentation at the valve level. The diagram below illustrates this structure, where blue nodes represent individual flare sources and all measurement is concentrated at the red collection points:

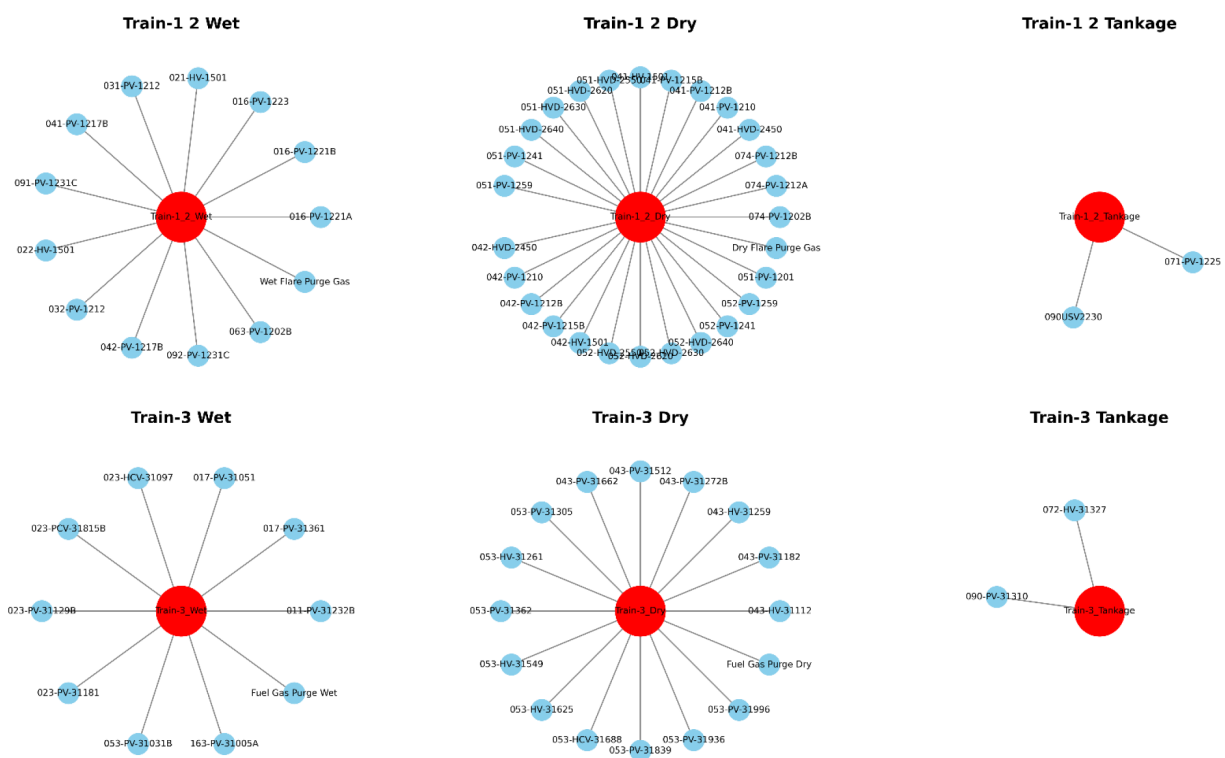


Figure 1—Flare network topology for Wet, Dry, and Tankage systems across LNG Trains 1/2/3

This header-only measurement approach introduces three critical blind spots:

1. **No attribution to individual sources**, limiting the ability to distinguish between operational necessities and process inefficiencies.
2. **No categorization of flaring events**, making it difficult to separate unavoidable safety flaring (e.g. continuous purge) from routine or failure-driven emissions.
3. **Delayed response and low accountability**, as root cause investigations often rely on manual reconciliation of trends, operator logs, and historian data—long after the flare event has passed.

Addressing these limitations requires a shift toward a data-driven diagnostic layer—one that can digitally reconstruct flare source behavior using existing process signals. Without requiring physical instrumentation upgrades, this strategy enables source-level quantification, categorical classification of flare events, and targeted intervention planning, all with the aim of achieving meaningful emissions reduction while safeguarding operational integrity.

Statement of Theory and Definitions

Effective flaring reduction hinges on the ability to distinguish between emissions that are fundamental to safe operations and those that are avoidable through better control, sequencing, or system design. Industry frameworks such as API Standard 521 (API Standard 521, 2014) and the OGMP 2.0 (OGMP 2.0, 2021) technical guidance establish foundational principles for classifying flaring into operationally meaningful categories. These categories are essential not only for reporting but also for designing targeted interventions.

In the context of LNG production, four distinct flaring classifications are adopted:

- **Safety Flaring:** Emissions required to maintain safe operating conditions, such as continuous purge gas to prevent air ingress and maintain inert systems. These are inherently necessary and typically considered non-reducible without compromising safety.

- **Routine Flaring:** Emissions that occur under normal operating conditions. While expected, they often reflect opportunities for optimization.
- **Non-Routine Flaring:** Emissions triggered by unplanned events including valve misoperation, control system anomalies, or impact due to ambient changes.
- **Start-Up / Shutdown Flaring:** Emissions occurring during plant transitions. These are influenced by start-up logic, valve sequencing, and fuel gas management practices.

At the Tangguh facility, the flare reduction program specifically targeted Safety, Routine, and Start-Up / Shutdown Flaring.

To quantify the minimum safe operating baseline for each category, the facility defines a concept known as the **Minimum Technical Limit (MTL)**. This represents the **lowest flaring volume achievable under stable, safe, and uninterrupted operation**—excluding flare events driven by equipment trips or abnormal conditions. MTLs are calculated based on historical low points during steady operation, adjusted for process reliability, redundancy needs, and safety margins.

This theoretical framework enables the separation of unavoidable emissions from those that can be minimized or eliminated. It also serves as the foundation for a structured flaring reduction strategy—where interventions can be planned not based on assumptions, but grounded in operational data, engineering judgment, and process safety principles.

Description and Application of Equipment and Processes

To effectively reduce flaring at scale, visibility must extend beyond totalized header data into the behavior of individual flare contributors. However, in large LNG facilities, instrumentation is typically concentrated at flare headers—limiting visibility to aggregate volumes while masking the behavior of specific valves or systems. Addressing this limitation requires a framework that can transform raw, unstructured process data into engineering insights, without relying on additional instrumentation.

To that end, the facility adopted a data-centric framework built on the DIKIW (Data–Information–Knowledge–Insight–Wisdom) hierarchy (Ackoff, 1989; Rowley, 2007). This approach systematically translated thousands of process signals into structured knowledge that enabled high-precision intervention planning. The framework integrates control valve analytics, operational classification, flow estimation modeling, and interactive dashboards to create a real-time flare decision support system.

From Data to Information: Mapping the Flare Landscape

The project began with 74 control valves connected to the Wet, Dry, and Tankage flare systems. Initially, these valves existed only as PI tags—labeled by numbers, disconnected from their operational context, and unlinked to any flaring behavior.

To convert this raw data into meaningful information, each valve was assigned a functional label. Using engineering drawings, operational experience, and historian trend analysis, the team mapped each tag to its process location and likely flare function. This mapping enabled the categorization of valves into four core flaring types:

1. **Safety-related** (e.g., continuous purge gas)
2. **Routine operational** (e.g., dryer switchovers, LNG loading to ship)
3. **Non-routine** (e.g., occurs due to unusual conditions not typically encountered during stable operation, such as pressure buildup caused by high ambient temperature)
4. **Start-up and shutdown** (e.g., full train depressurizations)

These classifications were visualized in a **node-link network diagram** where each valve is a node, color-coded by flare type as shown in Figure 2. This first transformation—from scattered tags to structured classification—established the informational foundation of the flare network.

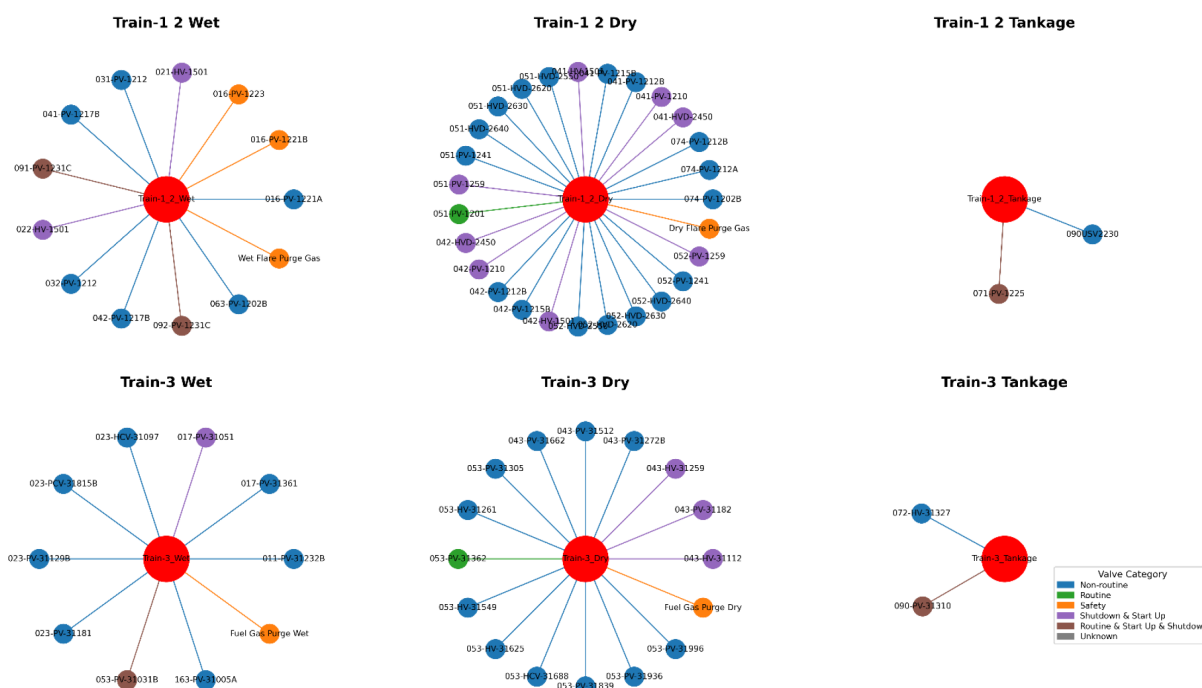


Figure 2—Categorized Valve Map by Flare Type

From Information to Knowledge: Grouping by Functional Clusters

Once the flare contributors were labeled, the next step was to convert this into system-level understanding. Valves were grouped into functional clusters—referred to as "flare rooms"—each representing a dominant flare category. For example, valves contributing to start-up emissions were grouped into one cluster, while purge-related valves were grouped into another.

This transformation revealed hidden patterns: some rooms were dominated by a few high-volume contributors, while others exhibited diffuse flaring from multiple small sources. The grouped visualizations allowed engineers to see systemic behaviors rather than isolated data points. Figure 3 shows the flare contributor grouping by its category.

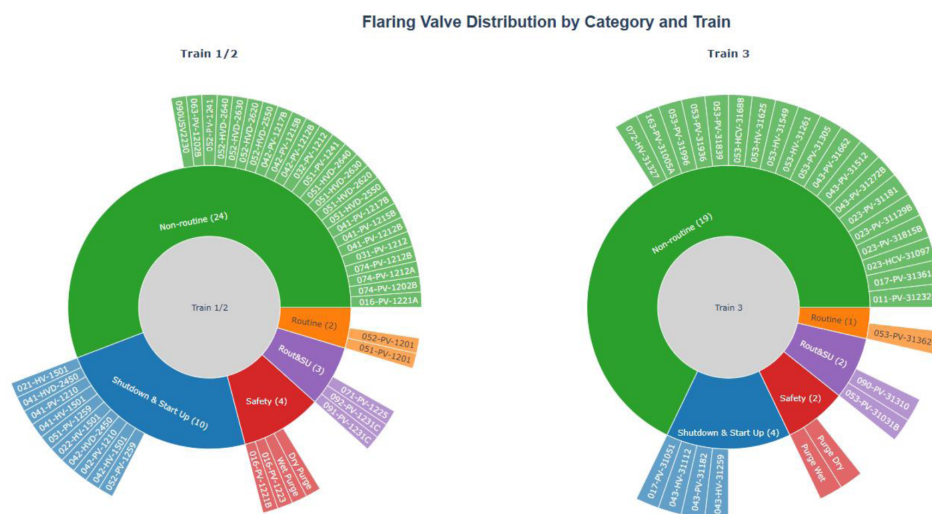


Figure 3—Flare Contributor Grouping by Category

At this stage, the team introduced the concept of the **MTL** — a theoretical baseline defining the lowest sustainable flare level that maintains safe operation. MTLs were defined per category and were grounded in historical low-flow performance during steady-state conditions.

From Knowledge to Insight: Estimating Valve-Level Flow Contributions

After establishing categorical groupings of flare sources, the next step was to quantify the magnitude of emissions at the equipment level. While classification provided a structural understanding of flaring contributors, it did not indicate the relative impact of each individual valve or event. This limitation hindered effective prioritization of engineering interventions and obscured the true drivers of flare load.

At Tangguh LNG, flow measurement devices are installed only at the header level—namely at the knock-out drums (KODs) of the Dry, Wet, and Tankage flare systems. These devices report cumulative volumes but cannot resolve individual contributions from the dozens of upstream valves feeding into each header. As a result, plant teams lacked the capability to determine which components within the network were the dominant sources of flaring.

To overcome this constraint, the team developed a digital system known as the Flaring Monitoring Dashboard, designed to estimate valve-level flow rates. By integrating process historian data, engineering calculations, and automated analytics workflows, the system delivers near real-time insights into source-level flaring—without requiring additional instrumentation or capital upgrades.

System Architecture and Modeling Approach. The flare estimation system is built on three core digital technologies:

1. PI System

As the data historian platform, PI System collects time-series data such as valve opening percentage, upstream and downstream pressure, and gas temperature. These parameters are recorded at one-day intervals for more than 74 flare-connected valves.

2. Palantir Foundry

Foundry serves as the analytics engine where Python-based workflows are developed within its Code Workbook environment ([Palantir Foundry Documentation, 2023](#)). These workflows apply the standard control valve equation to estimate flow:

$$Q = C_v \cdot \sqrt{\frac{\Delta P}{G}}$$

Where,

- Q = Estimated gas flow rate
- C_v = Valve flow coefficient, digitized as a function of valve opening percentage
- DP = Differential pressure across the valve
- G = Gas specific gravity

Each valve's C_v curve is digitized and scaled according to its operating characteristics. The system accounts for various valve types (e.g., linear, equal percentage) and adjusts the flow coefficient dynamically based on real-time stroke data. [Figure 4](#) shows a sample of C_v curve of the valve.

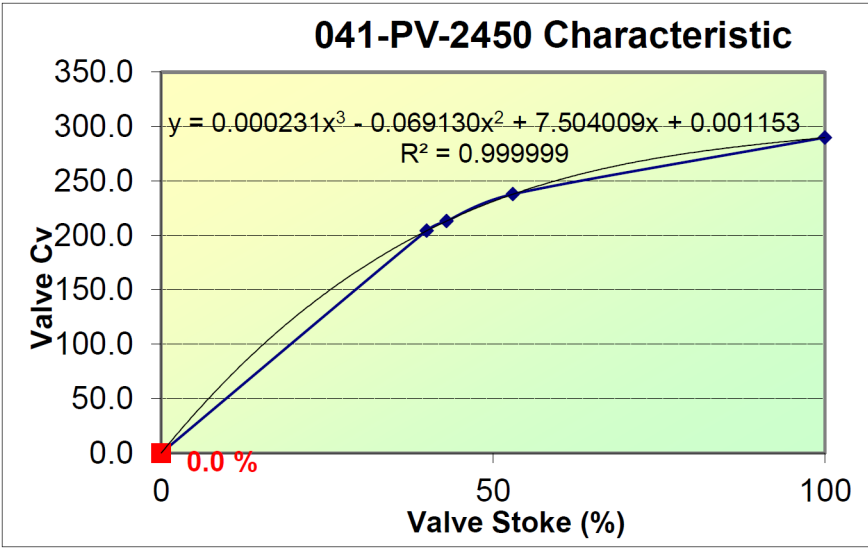


Figure 4—Sample Cv Curves Based on Valve Type

To implement this methodology, a Code-Workbook environment was configured in Palantir Foundry to enable automated processing. The system pulls both real-time and historical data for 74 flare-connected control valves, applying the flow calculation on a continuous time basis. For each valve, the estimated gas flow is derived using pressure differential, temperature, and valve opening percentage—adjusted according to the valve's specific flow characteristic curve. Figure 5 shows the Code-Workbook interface in the Palantir Foundry.

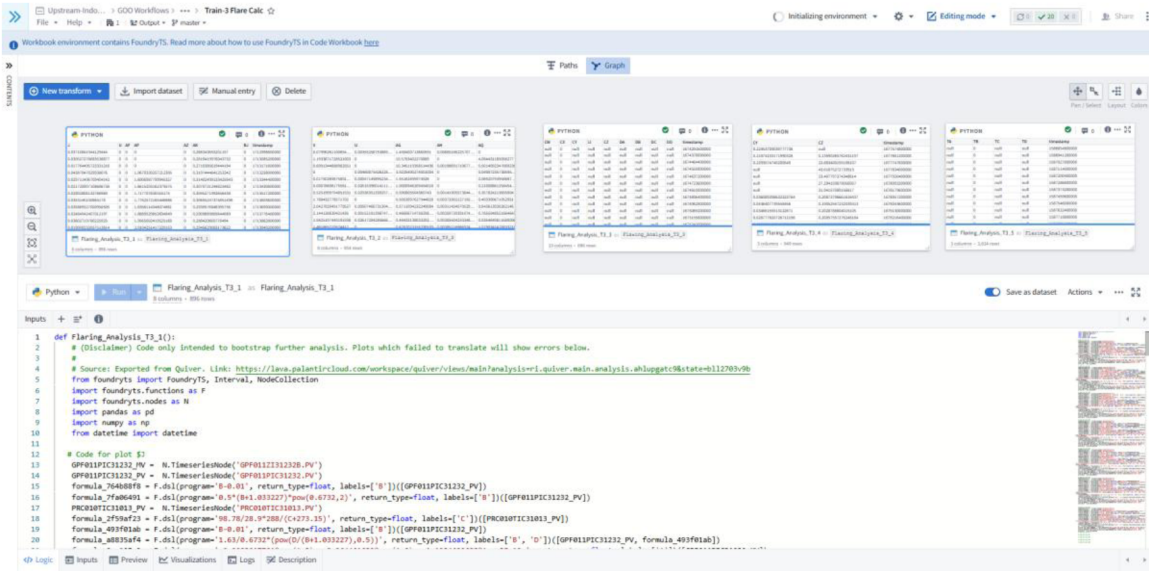


Figure 5—Code-Workbook Environment Interface in Palantir Foundry

The calculated outputs are then passed into Contour, a no-code data transformation tool in Foundry, where they are structured into standardized, time-stamped datasets suitable for visualization and trend analysis. Figure 6 shows the Contour interface in the Palantir Foundry.

Figure 6 shows the Palantir Foundry interface for 'Train3 Flaring Calculation Compiled'. The main view is a data table with 34 columns. The first few columns are 'Total_Flare', 'Tankage_Flare', 'Dry_Flare', 'Wet_Flare', and 'Timestamp'. The remaining columns are labeled '17PV1232', '17PV1233', '17PV1234', '23PV1237', '23PV1238', '23PV1239', '23PV1240', '23PV1241', '23PV1242', '23PV1243', '23PV1244', '23PV1245', '23PV1246', '23PV1247', '23PV1248', '23PV1249', '23PV1250', '23PV1251', '23PV1252', '23PV1253', '23PV1254', '23PV1255', '23PV1256', '23PV1257', '23PV1258', '23PV1259', '23PV1260', '23PV1261', '23PV1262', '23PV1263', '23PV1264', '23PV1265', '23PV1266', '23PV1267', '23PV1268', '23PV1269', '23PV1270'. The table contains 896 rows of data. Below the table, there are options to 'Export to' (CSV, JSON, etc.) and 'Export summary'. A 'Result' section at the bottom shows 'Train3 Flaring Calculation Compiled'.

Figure 6—Contour Interface in Palantir Foundry

Figure 7 depicts the data pipeline architecture that connects real-time PI data, automated flow calculations, and structured outputs. The integration ensures that flare data is refreshed regularly within a predefined time window, maintaining accuracy and consistency across all dashboards.

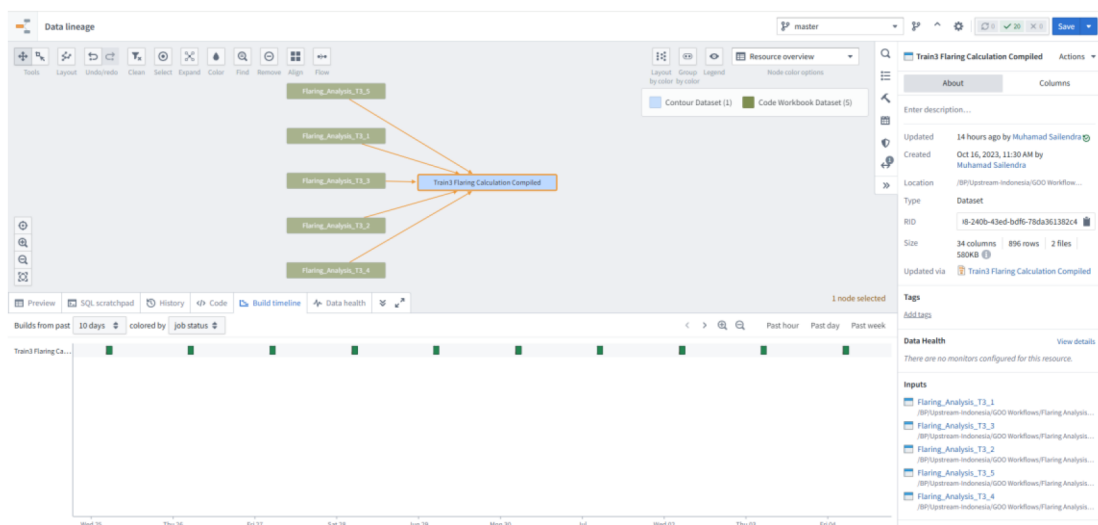


Figure 7—Data Lineage and Scheduling Flow Between Code-Workbook and Contour

3. Power BI

Processed data is visualized using Power BI, allowing end-users—including operators, engineers, and performance analysts—to interact with the flare data in real time. The dashboards provide functionality to:

- Monitor individual valve behavior and flare contribution
- Compare valve flow rates against calculated Minimum Technical Limits (MTL)
- Observe temporal patterns linked to operational changes
- Filter data based on processing train, and specific time ranges

Figure 8 shows the Power BI dashboard for Train-1/2.

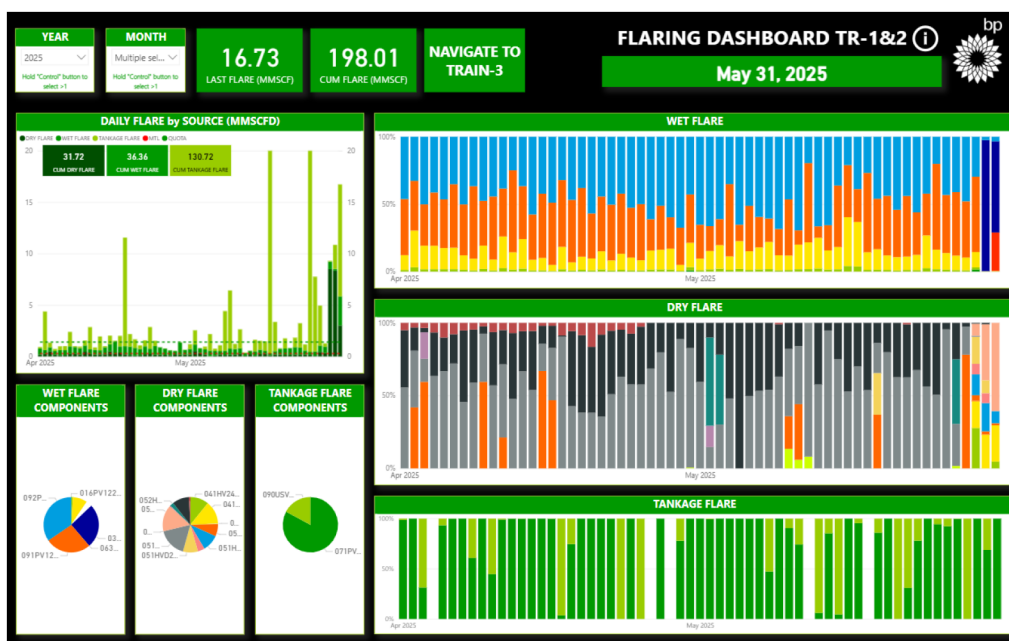


Figure 8—Flaring Monitoring Dashboard for Train-1/2

Figure 9 presents flare dashboard for Train-3. Users can explore trends, compare valve behavior between trains, and track the impact of operational interventions. The visualization supports proactive flare management and forms the basis for follow-up engineering analysis.

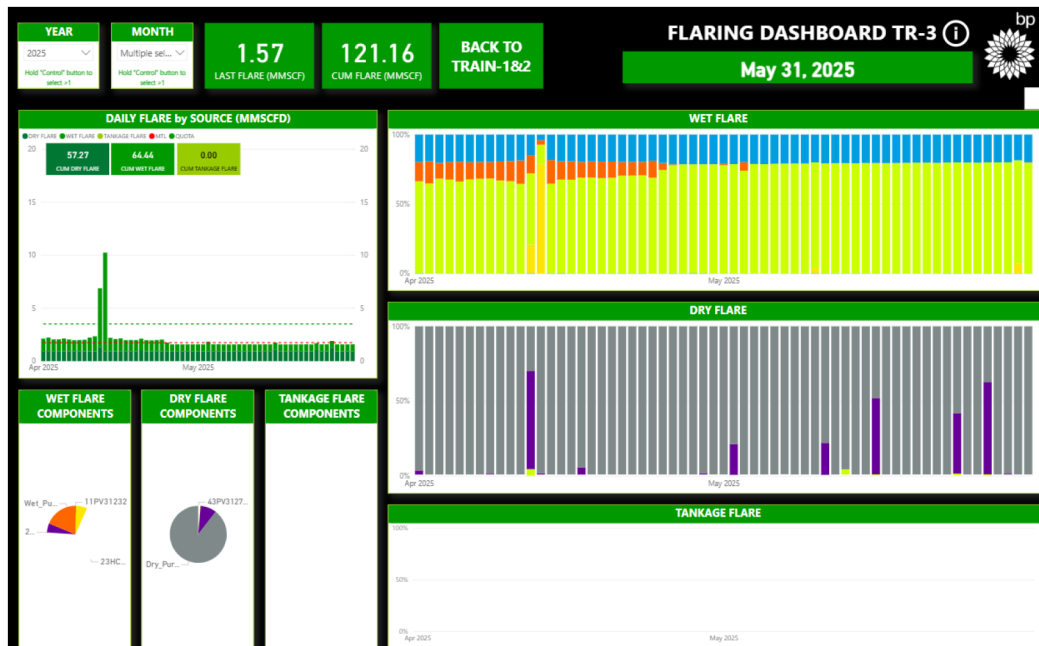


Figure 9—Flaring Monitoring Dashboard for Train-3

From Insight to Wisdom: Prioritizing High-Impact Contributors

Once valve-level flare flows were successfully modeled and visualized, the analytical framework progressed into its final layer: **Wisdom**. Within the context of DIKIW, this stage represents the capability to translate insight into strategic clarity—by isolating the most actionable contributors and aligning mitigation efforts

with operational priorities. The goal was not merely to describe system behavior, but to **support informed decisions** on where engineering or procedural interventions would yield the greatest impact.

The pivotal innovation at this stage was the **quantitative enhancement of the flare network visualization**. Building upon the categorical flare map established in earlier stages—where each node (control valve) was color-coded by flaring type (Routine, Safety, Non-routine, or Start-up)—a second dimension was introduced: **magnitude**.

Using the estimated flare rates derived from the modeling layer, each edge (pipe segment) connecting a valve to the flare header was dynamically weighted. **Line thickness** was scaled in proportion to the average volumetric contribution of that valve, calculated over a one-year period. This dual-coded representation allowed the network to communicate both the nature and the **severity of flare sources** in a single integrated view:

- **Node color** indicates flare classification,
- **Edge thickness** indicates contribution magnitude.

Figure 10 shows this enhanced flare network. The map combines categorical node coloring with magnitude-weighted edges, enabling users to quickly identify high-emitting valves—those whose connections appear significantly thicker than the surrounding network.

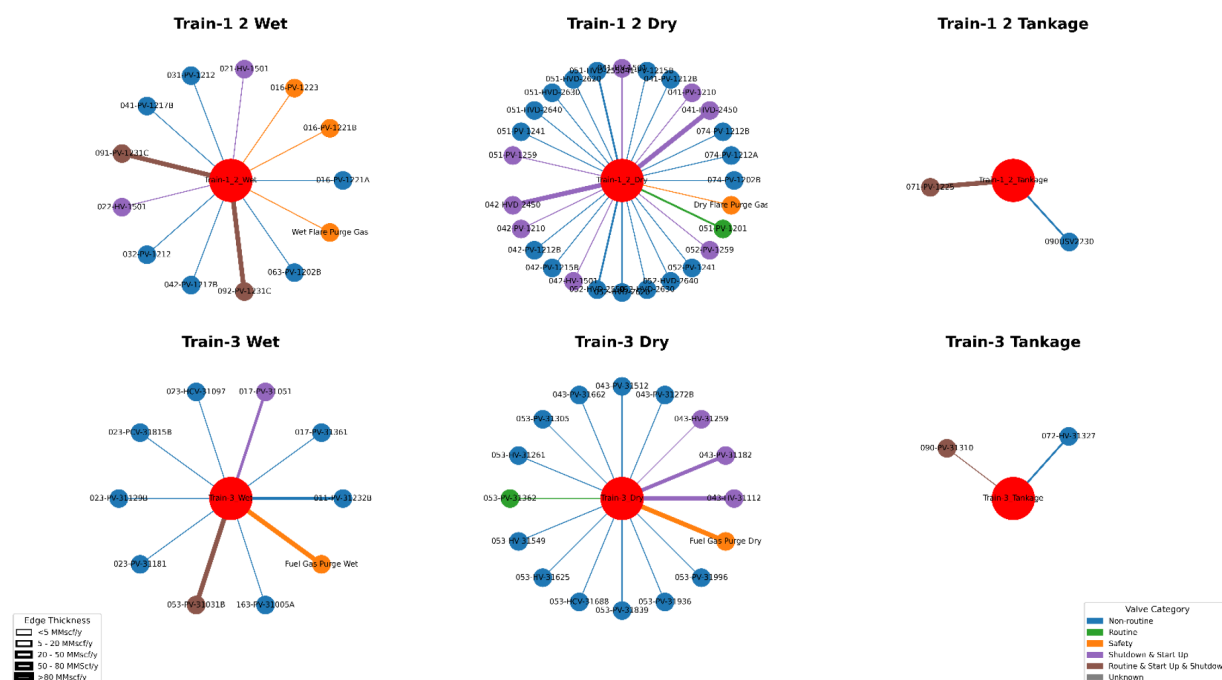


Figure 10—Enhanced flare network visualization incorporating flow estimation

This enhancement fundamentally shifted the map from a qualitative schematic to a **quantitative diagnostic tool**. Valves with unusually high flare emissions became immediately visible through disproportionately thick connectors, regardless of whether they belonged to critical safety systems or routine operations. These visual outliers—referred to as **flare "bad actors"**—emerged as high-priority candidates for engineering analysis, control logic review, or operator re-training.

To complement this network visualization, each valve's flare behavior was evaluated against its **Minimum Technical Limit (MTL)** — the minimum continuous flow required to maintain safe system integrity, such as header purging. This comparison enabled a strategic assessment of which valves exhibited **flare rates well above operational necessity**, offering low-regret opportunities for reduction.

By fusing **data modeling**, **visual analytics**, and **engineering thresholds**, this stage transformed flare management from a reactive task into a proactive, data-driven discipline. It laid the groundwork for the next phase: targeted mitigation planning and execution.

Flaring Analytics in Action: Findings, Priorities, and Strategic Roadmap

Flare Contribution by Operational Source

Figure 11 presents the breakdown of estimated flare volumes across the three LNG trains at Tangguh, grouped by operational source: Routine, Start-up, Safety, and Non-routine. The analysis is based on valve-level flow estimation over a defined monitoring period, with results aggregated by category.

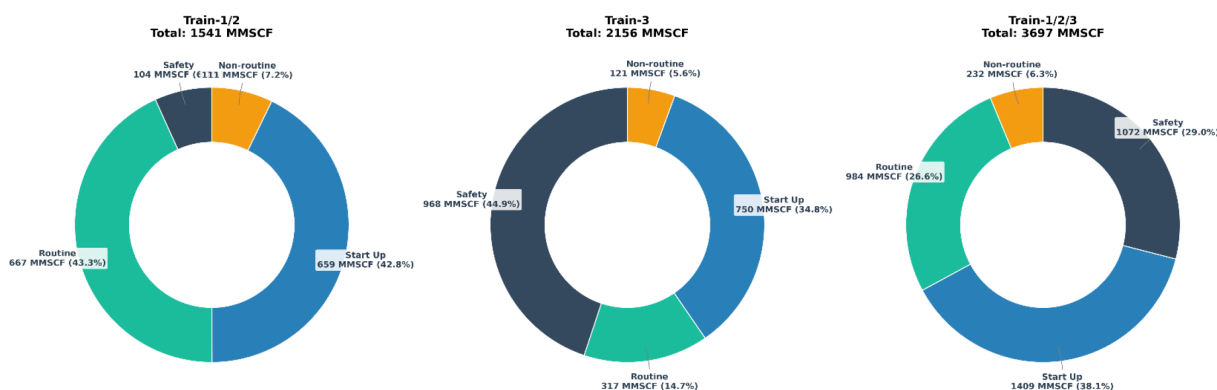


Figure 11—Contribution of Flare Sources by Operational Category

The flare contribution analysis reveals that emissions are not uniformly distributed across these categories. Instead, a substantial portion of flaring is concentrated in a few operational domains, indicating clear opportunities for focused mitigation:

- **Start-up and shutdown flaring** remains the largest contributor, totaling **1,409 MMSCF (38.1%)**. These emissions are typically associated with process depressurization, valve sequencing, and line clearing during transition phases such as plant start-up and cooldown.
- **Safety-related flaring**, primarily driven by continuous purge and sweep gas requirements, contributed **1,072 MMSCF (29.0%)**. While essential to maintain safe operating conditions and prevent air ingress, these flows reflect the necessary purge design rates.
- **Routine flaring** accounted for **984 MMSCF (26.6%)**, emerging from normal operational processes. Notably, two sub-systems present optimization potential:
 - **Dryer bed switchover**, where depressurization of spent beds prior to regeneration can be optimized through engineering modifications that allow for smoother pressure transitions or controlled pressure equalization, potentially eliminating associated flaring.
 - **LNG loading activities**, particularly **management of boil-off gas (BOG)** during ship loading transitions, represent another area where operational tuning or real-time balancing strategies can reduce unnecessary venting to flare.
- **Non-routine flaring** was the smallest contributor at **232 MMSCF (6.3%)**, but remains non-negligible. These emissions typically originate from **equipment trips**, **unexpected ambient or process condition changes**, or **operational parameters drifting outside optimum ranges**.

Flare Reduction Strategy and Technical Interventions

Figure 12 illustrates the projected reduction in flare volume across Tangguh LNG's three production trains, comparing current baseline emissions against optimized targets.

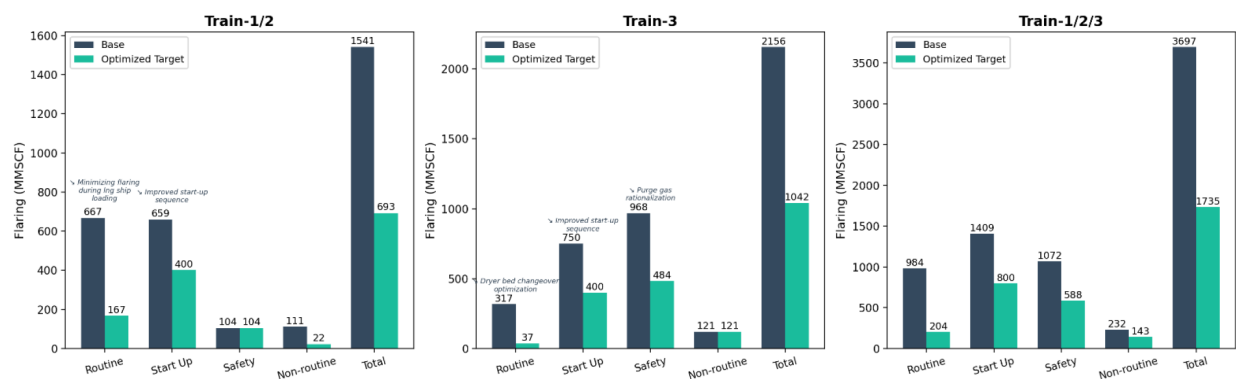


Figure 12—Estimated Flaring Reduction Potential by Operational Category

Through a combination of operational improvements and engineering interventions, total flaring can be reduced from **3,697 MMSCF/year** to **1,735 MMSCF/year**, representing an absolute reduction of **1,962 MMSCF/year**—equivalent to a **53% decrease in total flare emissions**. To achieve this, three primary intervention areas were identified:

1. Start-Up and Shutdown Sequence Optimization (*Potential reduction: 609 MMSCF/year*)

Transitional events such as unit start-up and shutdown were major contributors to peak flaring. Variability in execution between shifts and trains led to inconsistent outcomes. Mitigations included:

- Standardizing start up sequencing and timing based on historical low-flare benchmarks.
- Rolling out written procedures and targeted training sessions to ensure consistent execution plant-wide.

2. Purge Gas Rationalization (*Potential reduction: 484 MMSCF/year*)

Safety-related flaring, primarily from continuous purge gas, was identified as an opportunity for reduction. Key actions included:

- Validating new purge limits through internal engineering studies while maintaining compliance with API 521.
- Updating purge gas setpoints to reflect optimized purge flow without compromising system integrity.

This strategy led to significant reductions in steady-state flare loads, especially in Train-3.

3. Operating Parameter Optimization (*Potential reduction: 781 MMSCF/year*)

This strategy targets avoidable flaring during steady-state operations, particularly from **LNG ship loading** and **dryer bed changeover**. Interventions included:

1. LNG Loading Optimization

- Limiting vapor return flowrate during loading operations to minimize flare header overpressure.
- Pre-cooling the LNG rundown system prior to transfer to stabilize tank pressure and reduce BOG venting.
- Documenting and standardizing best practices from low-flare loading events across operational shifts.

2. Dryer Bed Changeover Optimization

- Extending depressurization duration during molecular sieve switchovers to reduce rapid pressure drop and gas discharge.
- Conducting fuel system assessments during dryer transitions, where excess HP fuel generated during switchover was redirected to LP fuel instead of being sent to flare.

These procedural changes required no equipment upgrades and delivered quick wins with cross-shift alignment.

Results: Verified Reduction and Progress Toward Target

Figure 13 illustrates the actual flare reduction achieved across Tangguh LNG's three production trains as of H1 2025, benchmarked against the baseline and optimized targets. Through targeted operational interventions, total flaring was successfully reduced from **3,697 MMSCF/year** to **2,556 MMSCF/year**, realizing an absolute reduction of **1,141 MMSCF/year**.

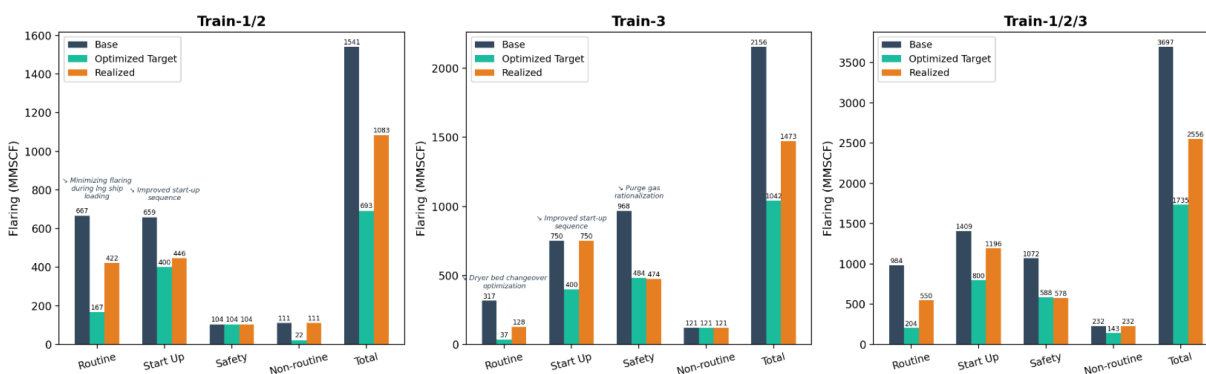


Figure 13—Realized Flaring Reduction vs. Baseline and Target

This corresponds to **58% progress** toward the full flare reduction target of 1,963 MMSCF/year.

These reductions were achieved through consistent deployment of identified optimization strategies:

- **Minimizing flaring during LNG ship loading** by controlling vapor return rates and pre-cooling LNG rundown lines before transfer operations.
- **Start-up sequence harmonization**, ensuring optimized start up sequencing and timing during plant transitions.
- **Routine operation improvements**, including:
 - *Extending depressurization duration during molecular sieve switchovers* to avoid rapid pressure drops and reduce gas discharge to the flare.
 - *Fuel system reassessment during dryer transitions*, enabling redirection of excess HP fuel to the LP fuel header instead of routing to flare.
- **Purge gas rationalization** by adjusting overly conservative purge flows in safety-critical valves without compromising system integrity.

Table 1—Summary of Realized Flare Reduction Performance

<i>Metric</i>	<i>Value</i>
<i>Baseline Flaring</i>	3,697 MMSCF/year
<i>Optimized Target</i>	1,735 MMSCF/year
<i>Realized Flaring (H1 2025)</i>	2,556 MMSCF/year
<i>Reduction Achieved</i>	1,141 MMSCF/year
<i>% of Target Achieved</i>	58% (from 1,963 MMSCF target)
<i>Economic Value Saved</i>	USD 7.99 million/year
<i>Emissions Avoided</i>	62,192 tCO ₂ e/year

Note: Economic value was estimated based on a reference market price of USD 8.2 per MMBtu for Tangguh LNG.

Conclusions

This paper presents a data-driven framework for flare source attribution and targeted reduction at Tangguh LNG. By integrating PI historian signals, engineering-based flow models, and operational classification into a centralized dashboard, the facility successfully transitioned from passive flare monitoring to an active, insight-driven decision support system.

The categorization of flare contributors into four types—Safety, Routine, Start-up, and Non-routine—provided a clear lens for intervention, while valve-level flow estimation uncovered specific contributors that were previously obscured by system-level measurements. This enabled targeted actions such as start up sequence optimization, LNG ship loading optimization, purge gas rationalization, and fuel system re-routing during dryer transitions.

As of H1 2025, these interventions have delivered a verified reduction of 1,141 MMSCF/year, equivalent to 58% of the total reduction target, translating to USD 7.99 million in value and 62,192 tCO₂e of avoided emissions. These results validate the practical applicability of the framework and mark a critical step forward in embedding emissions intelligence into everyday LNG operations.

Acknowledgement

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